

SRILEKHA MANDALI

Data Engineer | ETL Developer | Data Analyst

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SUMMARY

Data Engineer with hands-on experience designing, developing, and optimizing enterprise ETL pipelines, SQL-based data integration, and reporting solutions at Tata Consultancy Services. Currently completing an MS in Management Information Systems while contributing to technical research and writing roles and actively seeking to return to a full-time Data Engineer / Data Analyst / ETL Developer position. Skilled in Informatica PowerCenter, SQL, Python, and BI tools, with a proven record of improving pipeline performance and data quality across the full SDLC.

TECHNICAL SKILLS

ETL & Data Engineering: Informatica PowerCenter, ETL Pipeline Development, Data Integration, Data Transformation, SQL Optimization, Data Validation & Quality, Enterprise Data Warehousing

Programming & Query Languages: SQL (MS SQL Server, MySQL), Python, JavaScript, HTML, CSS

Analytics & Visualization: Power BI, Tableau, Looker Studio, Microsoft Excel (Pivot Tables, VLOOKUP/XLOOKUP), Dashboard Development, Data Visualization, Data Reporting

AI & Automation: Google Gemini AI, OpenAI API, LangChain, n8n, Make.com, Workflow Automation, Prompt Engineering

Data Processing & Integration: JSON Parsing, PDF Text Extraction, Structured Data Processing, API Integration

Enterprise Tools & Cloud: ServiceNow, JIRA, Autosys, Aqua Data Studio, MFT, Google Workspace APIs (Sheets, Docs, Gmail)

Methodologies: Agile Scrum, SDLC, User Acceptance Testing (UAT), Requirements Gathering, Business Process Analysis

WORK EXPERIENCE

Writing Assistant | Axcell LLC — Remote

Jul 2026 – Present

- Conduct technology research and analyze emerging trends for a technology management publication.
- Develop technical documentation, diagrams, and educational content in collaboration with subject matter experts.
- Organize technical references and synthesize research into structured, publication-ready documentation.

Graduate Teaching Assistant & Research Assistant | Auburn University at Montgomery (AUM) — Montgomery, AL

Aug 2025 – May 2026

- Assisted undergraduate and graduate students with coursework, technical guidance, and Gantt Project tutorials across online and face-to-face learning environments.
- Conducted research on entrepreneurial ecosystems and technology-business incubators using qualitative and quantitative research methods, contributing to academic publications, conference presentations, and poster sessions.
- Performed literature reviews, analyzed research data, and applied analytical methodologies to generate data-driven findings and recommendations, collaborating on projects related to Technostress, Innovation Ecosystems, and Incubator Success Factors.
- Served as Research Club Vice President, leading research-focused student initiatives, coordinating academic and professional development events, and collaborating with faculty to promote research engagement.

Systems Engineer | Tata Consultancy Services (TCS) — Hyderabad, India

May 2021 – Jun 2024

- Designed, developed, and maintained enterprise ETL pipelines using Informatica PowerCenter and SQL, integrating data from multiple enterprise source systems into centralized reporting environments supporting business operations.
- Designed and optimized SQL queries, ETL mappings, and data transformation workflows across 20+ pipelines, reducing query execution time by approximately 20% and enhancing overall ETL performance and reporting efficiency.
- Utilized Informatica PowerCenter, MySQL, Tableau, Python, Aqua Data Studio, Autosys, IDR, and MFT while managing development and production support activities through ServiceNow and JIRA, resolving an average of 30 tickets per month.
- Performed data validation and analyzed enterprise datasets of 2M+ records to identify data quality issues, ensuring accurate, reliable, and consistent reporting.
- Monitored, maintained, and optimized production ETL workflows by resolving workflow failures, infrastructure issues, permission errors, and performance bottlenecks, improving pipeline uptime/reliability by 15–20%.
- Collaborated with 5+ cross-functional stakeholders to gather business requirements and translate them into scalable ETL workflows and reporting solutions.

- Participated in the full Software Development Life Cycle (SDLC), including requirements analysis, development, testing, deployment, UAT, and post-production support within Agile Scrum teams.

Data Analyst | Roche — Mumbai, India

Jan 2021 – May 2021

- Analyzed operational and business datasets using SQL and Excel to identify trends and generate actionable insights.
- Developed Power BI dashboards and Excel reports to improve business visibility and support operational decision-making.
- Performed data extraction, profiling, cleansing, validation, and transformation to improve reporting accuracy and maintain high-quality datasets.
- Collaborated with stakeholders to gather reporting requirements and deliver analytical solutions that supported business objectives.

Software Engineer | Hexaware Technologies — Chennai, India

Jan 2020 – Jun 2020

- Developed a Canteen Management System using SQL, HTML, and CSS as part of core software development training.
- Participated in software testing, debugging, and deployment activities, and improved application performance and scalability through code optimization.

PROJECTS

AI Research Paper Summarizer and Organizer

Mar 2026 – May 2026

n8n, Google Gemini AI, LangChain, Google Sheets, Gmail, Google Drive

- Built a 27-node n8n workflow integrating Google Gemini AI via LangChain across six AI processing stages summarization, keyword extraction, research field classification, findings identification, difficulty rating, and audience recommendation with conditional routing.
- Automated end-to-end extract-transform-load of unstructured PDF research papers into structured, queryable records in Google Sheets, with automated Gmail notifications on completion.
- Diagnosed and resolved production issues including Gemini API rate limits (429 errors), service outages (503 errors), and duplicate-output bugs from merge misconfiguration through retry logic and corrected node expressions.

AI Portfolio Monitoring & Executive Analytics System

Mar 2026 – May 2026

n8n, OpenAI API, Google Sheets, Looker Studio, Workflow Automation

- Designed and implemented an end-to-end AI-powered portfolio monitoring system that automated financial news sentiment analysis, portfolio risk assessment, compliance monitoring, and executive reporting through integrated workflow automation.
- Built multiple n8n workflows to ingest portfolio data, evaluate sentiment trends, detect sector concentration risks, generate AI-driven investment recommendations, and synchronize results with Google Sheets and Looker Studio dashboards.
- Developed rule-based portfolio monitoring logic to identify constraint violations, portfolio exposure imbalances, and sentiment-driven actions (Hold, Review, Decrease), creating transparent and explainable decision support.
- Created interactive executive dashboards in Looker Studio featuring KPI monitoring, portfolio sentiment trends, sector distribution analysis, recommendation visualizations, and centralized portfolio monitoring tables to support data-driven decision making.

AI-Enabled Student Query Management & Announcement Automation

Mar 2026 – May 2026

Make.com, Google Gemini AI, Google Sheets, Gmail, Google Docs

- Built and deployed an AI-powered workflow for Auburn University at Montgomery that classifies student queries via NLP, generates draft responses, and routes high-priority cases for instructor/GTA review.
- Automated weekly announcement drafting and query status tracking, projected to cut manual communication handling time by 50%+.

PUBLICATIONS

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- “Building and Sustaining the Davao Innovation Ecosystem,” co-authored paper, 26th International Conference on Innovation-Driven Entrepreneurship and Finance, Manila, Philippines, 2026.
 - “Technology-Business Incubators Fuel the Growth in the Philippine Ecosystem: The Societal Impact,” co-authored paper, 28th Southern Association for Information Systems (SAIS) Conference, Asheville, NC, 2026.

EDUCATION

Master of Science in Management Information Systems (MSMIS) | GPA: 3.93/4.0

Auburn University at Montgomery (AUM) — Montgomery, AL

Aug 2024 – May 2026

Bachelor of Technology in Electronics and Communication Engineering | CGPA: 7.52/10

Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology — Chennai, India

Jun 2016 – Mar 2020